

## **Factors Affecting Fraud Detection Performance**

Considering the nature of the financial services industry, prevalence of fraud, and the negative effects of fraud, it is critical to assess, understand, and address all pertinent areas related to fraud and fraud detection. Key among the areas under consideration is an assessment of the factors affecting fraud detection performance. A combination of factors, as Hernandez Aros et al. (2024) observed, including class imbalance, evolving fraud strategies, concept drift, false positives, data quality, feature engineering, and model selection affect the effectiveness of fraud detection models and practices (Hernandez Aros et al., 2024). Further, in an evaluation and discussion of factors affecting the effectiveness of fraud detection, Vanini et al. (2023) and another report by Faisal et al. (2024) identified real-time processing requirements and computational efficiency as having a significant effect and impact on practical deployment of fraud detection models. Additionally, from another perspective, Zafar and Wu (2026) identified explainability and regulatory compliance as key emergent focus areas for financial institutions as they adopt AI in decision-making and fraud detection. Finally, Khatib (2024) identified institutional differences in key areas, such as transaction systems, customer behaviour, and data architectures affect model portability across banks. Evidently, from the reviewed literature, different authors and scholars have examined the different challenges independently, with very few studies investigating their collective influence on overall fraud detection models in fraud detection models and practices.

## **Class Imbalance and Techniques Used to Improve Model Performance**

One of the key challenges affecting model performance in fraud detection is class imbalance. Extreme class imbalance emanates from fraudulent transactions accounting for only a very small proportion of all the transactions in financial transactions in real-life and in datasets (Hashemi et al., 2022). Illustrating this challenge, Al-dahasi et al. (2025) observed that in most

cases, standard machine learning algorithms achieve high accuracy rates overall, but fail to identify minority fraud cases. To address the class imbalances, institutions and professionals have used different techniques and approaches, such as oversampling. According to Albalawi and Dardouri (2025) and Zhang et al. (2022), different oversampling techniques, such as SMOTE and ADASYN have helped professionals and institutions to improve minority class representation during model training, which finally improves fraud-detection in real-life. From another perspective, as Hashemi et al. (2022) observed, undersampling and cost-sensitive learning also improve fraud detection performance but may discard useful information or require careful parameter tuning, when used in cases of class imbalances. Finally, as Khalid et al. (2024) in one study and Al-dahasi et al. (2025) in a report observed, ensemble learning combined with imbalance mitigation consistently demonstrates higher precision, recall, and F1-score than single-model approaches. However, despite the significant improvements in addressing class imbalances and improving model effectiveness, it is evident that no imbalance mitigation technique has consistently outperformed others across different datasets and fraud-detection contexts.

### **Effects of Concept Drift, Evolving Fraud Patterns, and Adversarial Evolution**

Other notable challenges and issues affecting fraud detection effectiveness are concept drift, evolving fraud patterns, and adversarial evolution. As Adebayo et al. (2023) observed fraud and fraud patterns are evolving continuously because criminals modify their behaviours and endeavour to bypass current fraud detection systems, which reduces the effectiveness of static machine learning models. Adversarial evolution includes the numerous changes in fraud patterns and the intentions of criminals to attack and infiltrate systems and gain unauthorised access to customer data and resources (Hashemi et al., 2022; Khatib, 2024). Further illustrating the impact of evolving fraud and concept drift, Israfilov et al. (2025) observed that concept drift causes models that had been previously trained to lose their effectiveness in accuracy in

predicting fraud because of the changes in transactions and fraud patterns and characteristics. Therefore, with the continuous changes in transaction types and fraud patterns, the basis for models' training, change, which occasions concept drift.

Scholars, practitioners, and other stakeholders have leveraged and used different approaches to address concept drift, evolving fraud patterns, and adversarial evolution. For example, different research and studies have demonstrated that adaptive learning, online learning, incremental learning, and adaptive bagging foster resilience against evolving and emergent fraud strategies and techniques (Gupta & Babu, 2025; Goudar et al., 2026). Further illustrating the importance of real-time and adaptive learning, Khatib (2024) observed that in many instances, attackers purposefully and deliberately manipulate the transaction characteristics and details to evade ML classifiers, which requires adaptive and learning systems. Therefore, from the current literature, it is evident that fraud detection models are more reactive than proactive, which requires periodic and continuous retraining for continuous adaptation and learning for more responsive systems. Indeed, in line with these patterns, Gupta and Babu (2025) observed that it is important to ensure adaptive and self-learning fraud detection frameworks, which are capable of responding to dynamic fraud environments in real time.